

Spring Framework Annotations - Detailed Usage

Core Annotations

@Component

Marks a class as a Spring-managed component.

Usage:

@Component

```
public class MyService { }
```

@Service / @Repository / @Controller

Specializations of @Component to mark business, data, and web layers.

@Autowired

Injects dependencies by type.

Usage:

@Autowired

```
private MyService myService;
```

@Qualifier

Resolves ambiguity when multiple beans of same type exist.

Usage:

@Autowired

```
@Qualifier("myBean") private MyService myService;
```

Configuration and Beans

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@Configuration / @Bean

Defines beans in a Java config class.

Usage:

@Configuration

```
public class AppConfig {  
  
    @Bean  
  
    public MyService myService() { return new MyService(); }  
  
}
```

@Value

Inject values from properties.

Usage:

```
@Value("${app.name}") private String appName;
```

Spring MVC

@RestController

Combines @Controller and @ResponseBody.

@RequestMapping / @GetMapping / @PostMapping etc.

Maps HTTP requests to controller methods.

Usage:

```
@GetMapping("/hello")
```

```
public String hello() { return "Hello"; }
```

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`@RequestParam` / `@PathVariable` / `@RequestBody`

Bind query params, path variables, and request bodies.

Spring Boot

`@SpringBootApplication`

Main entry for Spring Boot app.

`@ConfigurationProperties`

Binds configuration to Java objects.

Usage:

`@ConfigurationProperties(prefix = "app")`

```
public class AppConfig { private String name; }
```

Security Annotations

`@EnableWebSecurity`

Enables web security.

`@PreAuthorize` / `@Secured`

Method-level security.

Usage:

`@PreAuthorize("hasRole('ADMIN')")`

```
public void deleteUser() { }
```

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Spring Data JPA

`@Entity` / `@Table` / `@Id` / `@GeneratedValue`

Used to map Java objects to DB tables.

`@Repository` / `@Query`

Custom queries and repository support.

AOP

`@Aspect` / `@Before` / `@After` / `@Around`

Used to define cross-cutting concerns.

Usage:

`@Aspect`

```
public class LoggingAspect {  
  
    @Before("execution(* pkg..*(..))")  
  
    public void log() { }  
  
}
```

Testing

`@SpringBootTest`

Loads full context for integration testing.

`@MockBean`

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Injects mock dependencies in Spring tests.